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# Real-time equivalence ratio measurements in gas turbine combustors with a near-infrared diode laser sensor

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**Abstract**

A rapid (2 kHz) near-infrared tunable diode laser (TDL) absorption sensor is developed for real-time measurements of methane concentration (and thus equivalence ratio) in gas turbine combustors. The non-intrusive sensor is based on fixed-wavelength laser absorption of methane near 1.65 μm using a single fiber-coupled diode laser. This $CH_4$ absorption wavelength is free of interference from the absorption of other species in air. Wavelength modulation spectroscopy (WMS) is combined with second-harmonic detection to improve the sensor sensitivity and accuracy. WMS-1*f* signal is used to normalize the 2*f* signal to remove the need for calibration and account for the laser transmission variations in gas turbine combustors. The near-IR TDL sensor is first calibrated in a static cell with $CH_4$ and $N_2$ mixture to determine the line-strength and find the laser setpoint. Test with a gas turbine combustor operating at atmospheric pressure is then conducted to demonstrate the sensor accuracy and response time. The TDL sensor is used to characterize equivalence ratio fluctuations in the combustor before the fuel/air mixtures enter the combustion chamber. The gain between heat release and equivalence ratio fluctuation is measured when the fuel flow rate is modulated at different levels. The combustion test results demonstrate the utility of the TDL sensor as a useful diagnostic tool to study flame characteristics and evaluate different combustor designs.
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*Keywords:* Diode laser; Wavelength modulation spectroscopy; Equivalence ratio measurement; Gas turbine combustor

## 1. Introduction

Lean premixed combustion has been widely applied to reduce pollutant emissions for gas turbine engines for both power and propulsions systems. In lean premixed flames, the dominant $NO_x$ formation mechanisms depend on the local flame temperature [1]. The gas turbine combustors that operate at lean fuel/air equivalence ratios significantly reduce $NO_x$ production due to lower flame temperatures. However, lean premixed combustion is susceptible to combustion instabilities in the form of thermoacoustic oscillations and lean blowout, which reduce efficiency, increase emissions, and pose significant safety hazards.

Thermoacoustic instability is self-sustained combustion oscillation near the acoustic frequency of the combustion chamber, which is the result of the closed loop coupling of unsteady heat release to pressure oscillations [2,3]. Intensive experimental and theoretical work has been performed to understand the driving mechanisms of

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thermoacoustic instabilities, and to suppress these instabilities in laboratory-scale and full-scale combustors [3–8]. It is well understood that heat release fluctuations can produce pressure oscillations; however, the mechanisms whereby pressure oscillations result in heat release fluctuations are not well known [4,6]. Equivalence ratio fluctuation is considered to be one of the most important driving mechanisms for thermoacoustic instabilities in fuel-lean gas turbine combustion systems [4–6]. Because of the complex physical and chemical interactions involved in thermoacoustic oscillations, it is difficult to predict this unstable combustion behavior. Typically, measured flame transfer function between equivalence ratio fluctuation and heat release fluctuation is required as the direct input to the analytic model. Therefore, measurement of equivalence ratio fluctuation during unstable combustion is of great importance to study and predict thermoacoustic instabilities in gas turbine combustor.

Equivalence ratio has been measured using infrared (IR) methane absorption of the 3.39 μm output of a He–Ne laser to study its effect on heat release during premixed unstable combustion [6]. Local fuel-to-air ratio was also measured by laser absorption at the same wavelength to study the effect of mixing on $NO_x$ emissions in premixed burner [9]. The same IR laser absorption technique has been used to measure fuel concentration in pulse detonation engines and internal combustion engines [10,11]. However, He–Ne lasers are sensitive to ambient conditions and simultaneously emit diffuse radiation and coherent light at multiple discrete wavelengths [12]. In addition, the absorption at 3.39 μm is the C–H asymmetric stretch bond common to all hydrocarbon fuels, while different hydrocarbons have different absorption coefficients. Therefore, the sensor needs to be calibrated for each fuel mixture encountered during operation. Thus, the IR absorption method has limitations for practical application in gas turbine engines.

In the present work, a rapid near-IR tunable diode laser (TDL) absorption sensor is developed for real-time measurements of methane concentration (thus equivalence ratio) in gas turbine combustors. The non-intrusive sensor is based on fixed-wavelength laser absorption of methane near 1.65 μm, and takes advantage of the large $CH_4$ mole fraction in unburned fuel/air mixture. This absorption wavelength is free of interference from the absorption of other species, and has been used for remote detection of methane leakage along pipelines [13,14]. Near-IR $CH_4$ absorption spectroscopic parameters have also been measured previously [15–17]. Advantages of this near-IR TDL sensor will be discussed compared to the previous sensors based on IR absorption at 3.39 μm. The TDL sensor is described, and calibrated in a static cell to determine the line-strength and find the laser setpoint. Test with a gas turbine combustor is then conducted to demonstrate the sensor accuracy and response time. The TDL sensor is used to characterize equivalence ratio fluctuations in the combustor before the fuel/air mixtures enter the combustion chamber. Gain between heat release and equivalence ratio fluctuation is measured with the TDL sensor when the fuel flow rate is modulated at different levels. To our knowledge, this is the first application of fixed-wavelength $2f/1f$ technique in gas turbine combustors for real-time equivalence ratio measurement.

## 2. Theory – wavelength modulation spectroscopy

The transmission coefficient $\tau(\nu)$ of monochromatic radiation through a uniform gas medium of length $L$ (cm) is given by the Beer–Lambert's law

$$\tau(\nu) = (I_t/I_0)_\nu = \exp[-P_i S \varphi_\nu L] \approx 1 - P_i S \varphi_\nu L, \quad (1)$$

where $I_t$ and $I_0$ are the transmitted and incident laser intensities, S ($cm^{-2}\,atm^{-1}$) and $\varphi_\nu$ (cm) are the line-strength and lineshape function for the absorption feature, and $P_i$ (atm) is the partial pressure of the absorbing species. The approximation in the right hand side holds for optically thin samples ($P_i S \varphi_\nu L < 0.1$).

Wavelength modulation spectroscopy (WMS), as an extension of absorption spectroscopy, is a well-known technique for improving the signal-to-noise ratio (SNR) [13,14,18–20] for TDL absorption measurements. In this technique, the laser wavelength is rapidly modulated:

$$\nu(t) = \bar{\nu} + a\cos(\omega t), \quad (2)$$

where $\bar{\nu}$ ($cm^{-1}$) is the center laser frequency and $a$ ($cm^{-1}$) is the modulation depth. For small modulation amplitude used in this work, the laser intensity modulation can be described by [18–20]

$$I_0(t) = \bar{I}_0[1 + i_0\cos(\omega t + \pi)]. \quad (3)$$

where $\bar{I}_0$ is the average laser intensity, $i_0$ is the intensity modulation amplitude. The transmission coefficient can be expanded in a Fourier cosine series:

$$\tau[\bar{\nu} + a\cos(\omega t)] = \sum_{k=0}^{\infty} H_k(\bar{\nu}, a)\cos(k\omega t), \quad (4)$$

and the second-harmonic Fourier component is given by

$$H_2(\bar{\nu}, a) = -\frac{S(T)P_i L}{\pi}\int_{-\pi}^{+\pi} \varphi(\bar{\nu} + a\cos\theta) \times \cos 2\theta d\theta. \quad (5)$$

For WMS detection, a lock-in amplifier is used to measure the second-harmonic ($2f$). This technique

is sensitive to absorption line shape curvature and is insensitive to low frequency noise [20]. Thus, WMS-2*f* detection offers benefits over direct absorption in terms of noise resistance and sensitivity. The lock-in amplifier acts as a band-pass filter and rejects noise outside the lock-in bandwidth. In addition, WMS measurements eliminate the need of problematic baseline fitting required in scanned-wavelength direct absorption measurements, especially in the case of weak absorbance. Finally, normalization of the WMS-2*f* signal with the 1*f* signal removes the need for calibration and account for the laser transmission variations due to beam steering, scattering, and window fouling [18–20]. Such normalization is important for gas turbine applications with intrinsic laser transmission variations. The magnitude of the WMS-2*f* and -1*f* signals can be written as [20]

$$S_{2f} = \frac{G\bar{I}_0}{2}\left|H_2 - \frac{i_0}{2}(H_1 + H_3)\right|,$$
$$S_{1f} = \frac{G\bar{I}_0}{2}\left|H_1 - i_0(H_0 + \frac{H_2}{2})\right|, \quad (6)$$

where G accounts for the optical-electrical gain of the detection system and laser transmission losses.

In our TDL sensor design, fixed-wavelength WMS with second-harmonic detection is used to simplify data processing and facilitate real-time equivalence ratio measurement. The laser wavelength is set at the line center of the absorption feature to maximize WMS-2*f* signal, where the 1*f*-normalized WMS-2*f* signal can be simplified as [20]:

$$R = \frac{S_{2f}}{S_{1f}} = \frac{H_2}{i_0}$$
$$= \frac{S(T)P_iL}{\pi i_0}\int_{-\pi}^{+\pi}\varphi(\bar{\nu} + a\cos\theta)\cos 2\theta d\theta. \quad (7)$$

The measured ratio $R$ is directly proportional to the partial pressure of absorption species if the line shape function does not vary with test conditions. If the gas pressure and temperature are measured, the equivalence ratio can be determined from the measured ratio R with known fuel composition. Without knowing temperature and pressure, equivalence ratio fluctuation ($\varphi'/\varphi$) can be determined from measured $R'/R$.

## 3. Sensor design

Figure 1 shows the simulated methane absorption spectrum (with HITRAN 2004 database [21]) for typical lean premixed gas turbine conditions (1 and 15 atm, 6% $CH_4$ in air, $L = 7$ cm, 700 K). Previous He–Ne laser based methane concentration sensors utilized IR absorption of the fundamental C–H stretch vibrational mode ($v_3$) at 3.39 μm [6,10]. In this work, the real-time TDL sensor

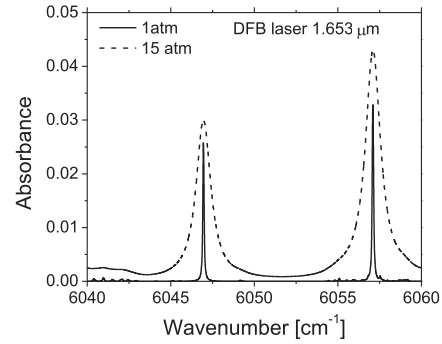


Fig. 1. Simulated absorption spectra for $CH_4$ absorption. Six percent $CH_4$ in air, $L = 7$ cm, 700 K.

for equivalence ratio is based on near-IR absorption of methane around 1.65 μm ($2v_3$ band). It is shown that there is minimum interference for methane absorption measurements at both 1.65 and 3.39 μm from other species in air [22]. The R(3) line around 1653.725 nm is well isolated at atmospheric pressure, and thus is selected for our TDL sensor. This line consists of three individual absorption lines with same lower state energy [22].

There are several advantages of this TDL sensor compared to previous sensors based on direct absorption technique [6,9–12,23]. First, the near-IR fiber-coupled TDL sensor takes advantage of mature technology for distributed-feedback (DFB) telecom diode lasers, fibers and optics, and thus has much higher SNR. Second, the diode laser output wavelength and power are very stable, and the laser wavelength is readily tuned with temperature and injection current. WMS can be combined with second-harmonic detection to improve the TDL sensor sensitivity and accuracy [24]. Third, the TDL sensor is not sensitive to transmission loss from window fouling or beam steering by using 1*f*-normalization. The $CH_4$ absorption level at near-IR is more suitable for line-of-sight applications to ensure sufficient transmission and absorption. The absorbance near 1.65 μm stays around 3–4%, which is ideal for WMS-2*f* measurement.

Figure 2 shows the schematic of the real-time TDL sensor for equivalence ratio in gas turbine combustors. The fiber-coupled DFB laser (NEL NLK1U5FAAA) operating near 1653 nm is mounted in a commercial laser mount (ILX Lightwave LDM-4980) and maintained at constant temperature (ILX Lightwave LDM-3900). The laser wavelength is modulated by a 10 kHz sinusoidal current waveform to generate modulation depth of 0.08 $cm^{-1}$, which is inferred from the interference pattern produced by a Ge etalon with a free spectral range of 2 GHz. The laser beam is divided into two paths by the fiber splitter. For the first path, the light is collimated into free space

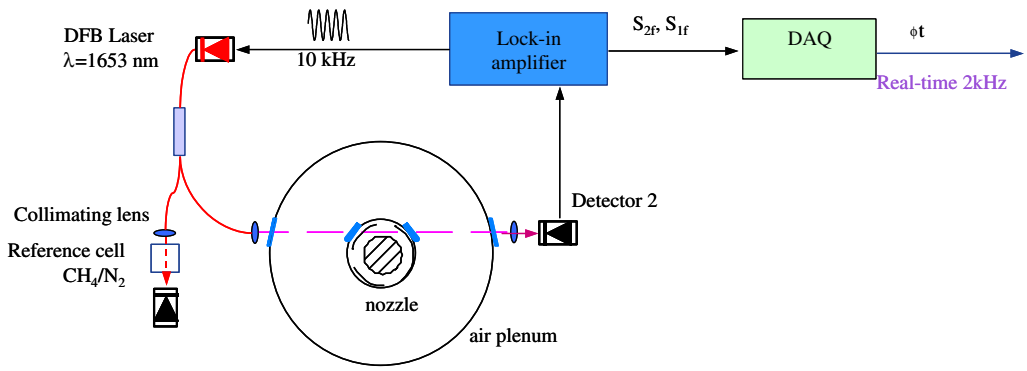


Fig. 2. Schematic diagram of the real-time TDL sensor for equivalence ratio in gas turbine combustors.

and transmitted through a static cell filled with calibration gas ($CH_4$ and $N_2$ mixture). For the second path, the light goes through a set of 25 mm sapphire windows which are mounted flush with the walls on the air plenum and gas turbine combustor. The laser beam is thus not perpendicular to the windows to minimize optical etalons. The laser beam is then focused onto a Ge detector (Thorlabs PDA50B). The free space light paths are exposed to air since there is negligible interference absorption. The detector signal is demodulated by a Perkin–Elmer lock-in amplifier (model 7280) to simultaneously recover the 1*f* and 2*f* signals with a time constant of 0.5 ms. The sensor bandwidth may be improved by using two lock-in amplifiers or a software lock-in [19,20]. A fast PC equipped with NI-DAQ board is used to record signals and provide real-time equivalence ratio output at 2 kHz.

## 4. Static cell measurement

Before the near-IR TDL sensor is applied to make measurements in gas turbine combustors, it is first characterized with a well-controlled environment, an atmospheric pressure static cell with a mixture of 5.22% $CH_4$ in $N_2$ (2% uncertainty), to determine the line-strength and find the laser setpoint.

### 4.1. Direct absorption measurement

For direct absorption measurement, the laser wavelength is tuned with a linear current ramp at a frequency of 100 Hz. Figure 3 illustrates the measured methane absorption line shape around 1653.725 nm in the static cell (1 atm, 297 K, $L = 4.5$ cm). The experimental profile is best-fit using a Voigt profile, and the residual (difference between data and fit normalized by the peak absorbance) is shown in the upper panel. A similar methane absorption lineshape is observed in atmospheric pressure gas turbine combustor running with non-preheated air. The line-strength for this absorption feature is inferred to be 0.0816 $cm^{-2}$/atm using the

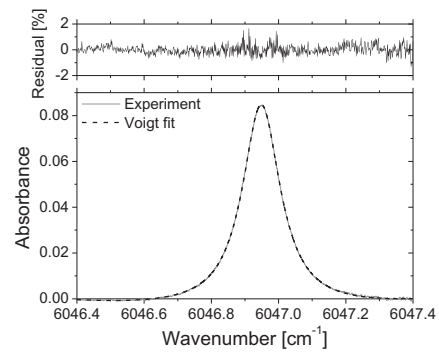


Fig. 3. Measured absorption spectrum in the static cell with 5.22% $CH_4$ in $N_2$. 1 atm, 297 K, $L = 4.5$ cm.

integrated absorbance area, which is about 4.9% higher than the HITRAN value. The measured value is used in the following test.

### 4.2. WMS measurement

For WMS measurement, the laser wavelength is sinusoidally modulated at 10 kHz. Figure 4 shows the measured WMS-1*f* and -2*f* spectra when the

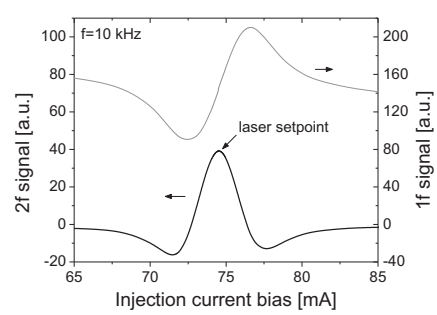


Fig. 4. Measured WMS-1*f* and -2*f* signals for the absorption feature around 1.653 μm in the static cell with 5.22% $CH_4$ in $N_2$. Laser modulated at 10 kHz. 1 atm, 297 K.

injection bias current is varied from 65 to 85 mA. The 1*f* signal has a large offset due to amplitude modulation (see Eq. (6)). Since the 2*f* signal peaks at the absorption line center, the laser wavelength can be manually locked by maximizing the 2*f* signal for fixed-wavelength WMS measurement. This process is repeated before each combustion test to ensure correct laser setpoint. We did not observe noticeable drift during normal 4-h test period. However, a small drift is observed over a long term (1 month) for the laser used in this work.

## 5. Measurements in gas turbine combustor

The TDL sensor is used to measure equivalence ratio fluctuations in a dry low $NO_x$ gas turbine combustor to study the flame characteristics. The atmospheric pressure, swirl-stabilized gas turbine combustor is fueled with natural gas and is shown schematically in Fig. 5. The flow is from left to right. The natural gas and air flow rates are controlled by valves and measured with calibrated flow meters. The measurement uncertainty of the equivalence ratio is estimated to be 0.01. A siren device which consists of a stator and a rotor is used to modulate the air flow up to 450 Hz [25]. The air enters the plenum which houses the gas turbine combustor. The fuel flow rate can be modulated by an electronic valve up to 300 Hz. The natural gas is delivered to the end cover of the combustor and is injected into the air flow through the holes on the swirler vanes.

The laser beam of the TDL sensor is lens-collimated and directed across the combustor right before the fuel/air mixture enters the combustion chamber. Cold flow mixing test with a sampling probe is conducted to measure methane concentration in the cross section of the laser beam location. Results show that the fuel/air mixture is relatively uniform in the cross section with premix efficiency ($=1-\sigma/\mu$, where $\sigma$ and $\mu$ are the standard deviation and mean of fuel concentration) >95% for typical operating conditions ($\varphi = 0.62$, air 0.177 kg/s, 297 K). The combustion chamber is bounded by a 50 cm long quartz tube ($D = 25.4$ cm), which provides optical access for flame visualization. CH*, OH*, $CO_2$* chemiluminescence signals are detected to quantify overall heat release fluctuations. It is assumed that the fractional change in heat release is equal to the fractional change in chemiluminescence signal [6,25]. The light emitted by the whole flame is collected by a fused silica fiber, split into several paths, and then filtered by a 10-nm band-pass filter and detected with a photomultiplier tube for each species [26].

### 5.1. Steady combustion

Combustion tests with steady conditions is conducted to demonstrate the TDL sensor accuracy and response time. Figure 6 shows the measured WMS-1*f* and -2*f* signals when siren is not operating ($\varphi = 0.65$). The measured 1*f*-normalized WMS-2*f* signal $R$ is also shown in Fig. 6. A time resolution of 0.5 ms is achieved for the TDL sensor with a lock-in time constant of 0.5 ms. The measured ratio is not steady, probably due to unsteady turbulent mixing. Figure 6 also shows the fast Fourier transform (FFT) spectrum calculated for the 0.5-s series of data. No distinct frequency is found on the FFT spectrum, indicating random equivalence ratio fluctuation along the sensor line-of-sight.

Figure 7 shows the measured WMS-2*f*/1*f* ratio $R$ as a function of equivalence ratio for fixed air flow rate. The bar in the figure represents the measured ratio fluctuations for different test conditions. The simulation result using Eq. (7) is also shown in Fig. 7 for comparison. One point calibration at $\varphi = 0.65$ is used in simulation to find the exact path length in the combustor. In all, the TDL sensor measurement is in excellent agreement with model. The measured ratio $R$ is linear with equivalence ratio for the tested conditions. The sensitivity ($dR/R/d\varphi/\varphi$) is calculated to be 0.9 instead of 1, mostly due to other species in natural gas (94.22% $CH_4$, 3.16% $C_2H_6$, balance with $N_2$ and $CO_2$): $\varphi = 9.4962–X_{CH_4}/(0.9422–X_{CH_4})$.

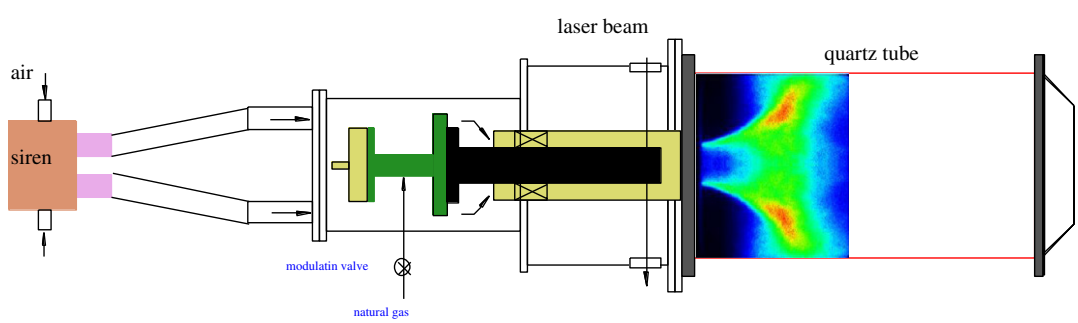


Fig. 5. Schematic of the atmospheric pressure test rig with a gas turbine combustor.

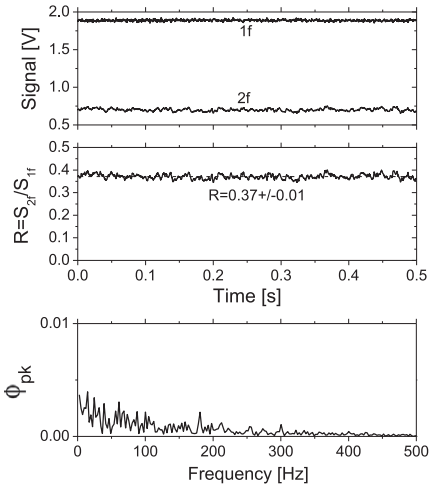


Fig. 6. Measured 1*f* signal, 2*f* signal, and ratio $R = S_{2f}/S_{1f}$ in the gas turbine combustor during steady combustion. $P = 1$ atm, $T = 297$ K, air = 0.177 kg/s, $\varphi = 0.65$.

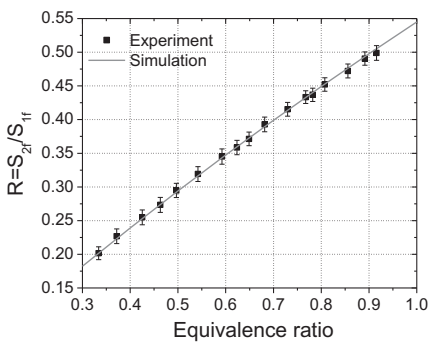


Fig. 7. Measured ratio $R$ as a function of equivalence ratio in the gas turbine combustor with steady combustion. $P = 1$ atm, $T = 297$ K, air = 0.177 kg/s.

### 5.2. Air-forced flame

The TDL sensor is then used to characterize equivalence ratio fluctuations in the gas turbine combustor when airflow is acoustically forced and modulated by the siren. Since the fuel flow is fixed at constant, the equivalence ratio near combustor exit is changing due to varying air flow rate when siren is operating at low frequency. Figure 8 illustrates the measured time history of WMS-1*f* and -2*f* signals, and equivalence ratio for the forcing frequency of 42 Hz (mean $\varphi = 0.63$). It can be seen from the 1*f* signal that the laser transmission signal is fluctuating, probably due to the vibrations of windows and test rig introduced by siren. This result clearly demonstrates the advantage of the TDL sensor with automatic laser transmission correction using 1*f*-normalization. Figure 8 also shows the FFT spectrum calculated from the measured

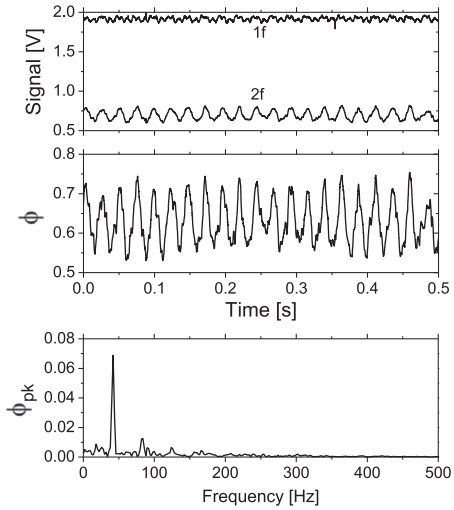


Fig. 8. Time history and FFT spectrum of the measured equivalence ratio when air flow is forced at 42 Hz. $P = 1$ atm, $T = 297$ K, air = 0.177 kg/s.

equivalence ratio data. The FFT is performed on 0.5-s of TDL sensor data, providing a resolution of 2 Hz. The dominant oscillation mode (42 Hz) and its harmonics can be clearly seen from the FFT spectrum. The zero-to-peak equivalence ratio oscillation of the fundamental mode is about 0.069. Without 1*f*-normalization, the peak equivalence ratio oscillation would be 0.075 (9% higher than actual value). This result demonstrates that the rapid TDL sensor can be used to accurately characterize equivalence ratio fluctuations in gas turbine combustor.

### 5.3. Fuel-forced flame

The rapid TDL sensor is also used to measure flame transfer function between heat release ($q$) and equivalence ratio fluctuation when the fuel flow is modulated by the electronic valve (air flow kept constant). The transfer function data characterize the sensitivity of flame in response to equivalence ratio fluctuation, and provide direct input for the analytical model to predict thermoacoustic instability during other operating conditions. Figure 9 shows the measured time history of equivalence ratio and CH* chemiluminescence for mean equivalence ratio of 0.59. The heat release oscillation is seen to be nearly in phase with the equivalence ratio fluctuation, where the zero-to-peak equivalence ratio fluctuation is 0.11 (19% of $\varphi_{\text{mean}}$) and the zero-to-peak heat release fluctuation is 0.12 (53% of $q_{\text{mean}}$). In addition, there is clear evidence of high harmonics in the heat release fluctuation which are less apparent in the equivalence ratio fluctuation. The FFT is performed on 7-s of TDL sensor data.

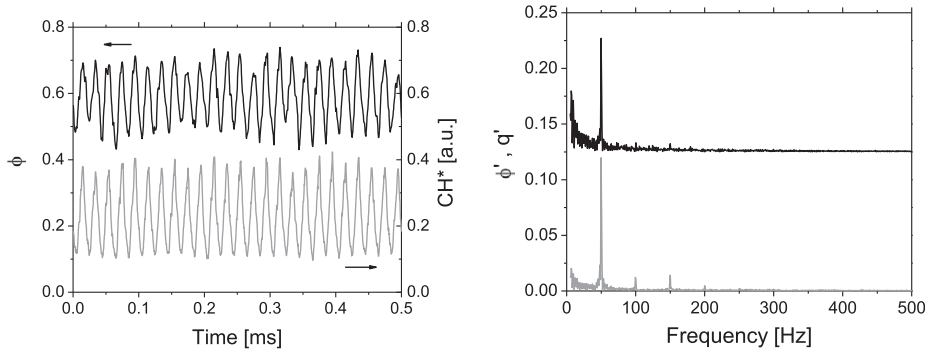


Fig. 9. Time traces and FFT spectra of CH* chemiluminescence and equivalence ratio when fuel flow is modulated at 50 Hz. $P = 1$ atm, $T = 578$ K, air = 0.10 kg/s.

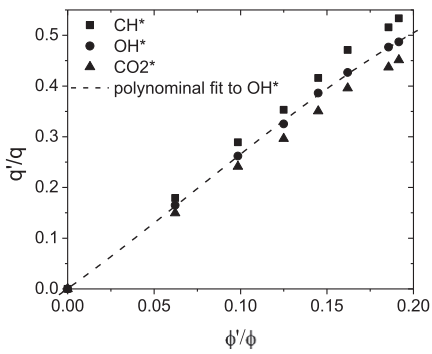


Fig. 10. Measured chemiluminescence vs. equivalence ratio fluctuation when fuel flow is modulated at 50 Hz. $P = 1$ atm, $T = 578$ K, air = 0.10 kg/s.

It is important to study flame response to different level of equivalence ratio fluctuations. The fuel flow modulation amplitude is controlled by the supply modulation voltage to the electronic valve. The measured overall heat release as a function of equivalence ratio fluctuation is plotted in Fig. 10. Note CH*, OH* and $CO_2^*$ chemiluminescence provide similar behavior, and thus can be used as an indicator of global heat release for studying flame response to equivalence ratio fluctuations [6]. A polynomial fit to the OH* data is also shown in Fig. 10, indicating the flame response starts to roll over at high equivalence ratio fluctuations ($\varphi'/\varphi > 15\%$). These results clearly demonstrate the utility of the TDL sensor as a useful diagnostic tool to study combustion instabilities.

## 6. Conclusions

A rapid (2 kHz) TDL absorption sensor has been developed for real-time measurements of equivalence ratio in gas turbine combustors. The non-intrusive sensor is based on fixed-wavelength laser absorption of methane near 1.65 μm using single fiber-coupled diode laser. This wavelength is free of interference from the absorption of other species in air. Wavelength modulation spectroscopy is combined with second-harmonic detection to improve the sensor sensitivity and accuracy. The WMS-1*f* signal is used to normalize the WMS-2*f* signal to remove the need for calibration and account for the laser transmission variations due to beam steering, window fouling, and vibration. The near-IR TDL sensor offers some advantages for practical gas turbine applications over previous sensors based on direct absorption technique. First, the near-IR methane sensor takes advantage of mature technology for telecom lasers, fibers and optics, and thus has better SNR. In addition, the near-IR TDL sensor is insensitive to transmission loss from window fouling or beam steering.

The near-IR TDL sensor is first calibrated in a static cell with $CH_4/N_2$ mixture to determine the line-strength and laser setpoint. Tests with a swirl-stabilized gas turbine combustor are then conducted to demonstrate the sensor accuracy and 2 kHz response time. The TDL sensor is used to characterize equivalence ratio fluctuations before the fuel/air mixtures enter the combustion chamber. Gain between heat release and equivalence ratio fluctuation is measured when the fuel flow rate is modulated at different levels. These combustion test results demonstrate the utility of the TDL sensor as a useful diagnostic tool to study combustion instabilities and evaluate different combustor designs.

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